Group 31: Taehoon Kim(tkim363), Diego De Abreu (ddea182), Harrison He (zhe909)

# COMPSYS 305: Mini Project

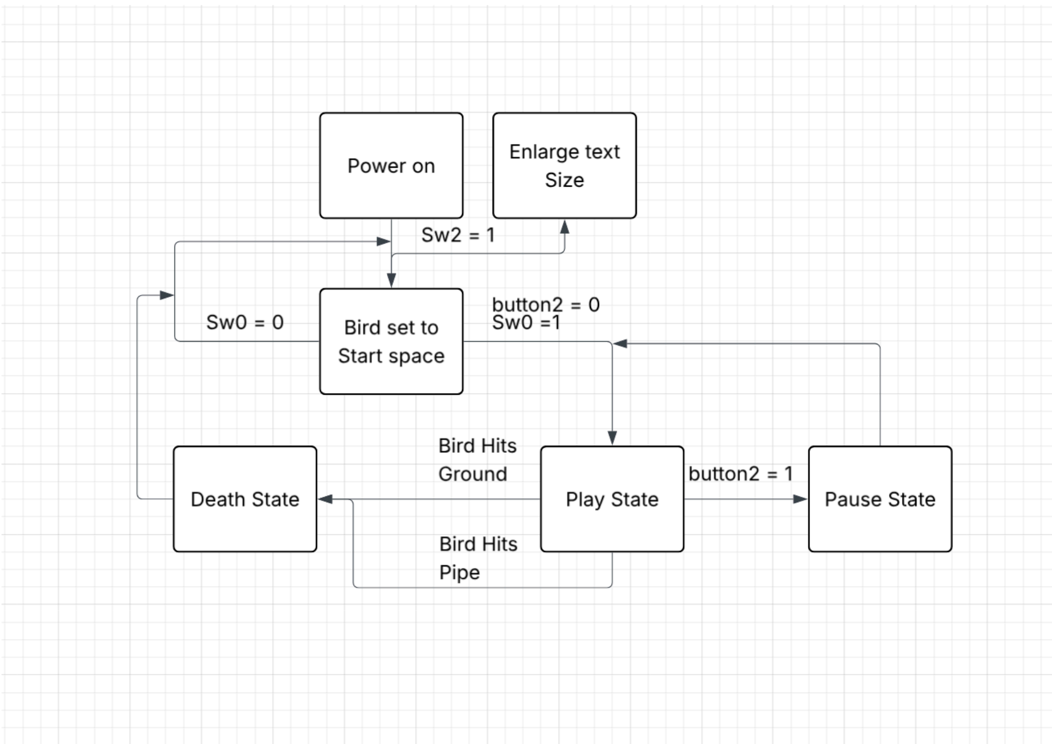
## Design Report

# 1. Game Strategy

Our project implements a real-time hardware "Flappy Bird" on the DE0-CV FPGA, focusing on hardware-level physics and VGA synchronization.

- **Vertical Physics**: Gravity is simulated by increasing the per-frame vertical velocity ($v_y$). When the PS/2 mouse clicks, an upward impulse is applied, resulting in a "flap" behavior.
- **Collision Detection**: Overlapping real-time ($x$, $y$) coordinate comparisons between birds and pipes/grounds trigger a GAME_OVER state.
- **Difficulty Scaling:** To meet project requirements, adjust the speed and pipe spacing height dynamically across three steps based on the current score.
- **LFSR Randomization**: Linear Feedback Shift Register (LFSR) provides pseudo-random pipe heights for different gameplay.
- **VGA Synchronization**: All logic updates occur during V_Sync to ensure flicker-free rendering.

# 2. Block Diagram



# 3. State Machine

Our team's game control logic is designed with a Moore-style Finite State Machine (FSM) for hardware efficiency. The entire system has four main states, each of which is switched by the user's input (button, mouse) and in-game signals (collision detection).

## i. State Definitions

### a. IDLE (Initial Screen): The state of the system being powered on or immediately after reset.

- Screen output: A standby message such as "Press Start" or "Hello World" is displayed on the screen.
- Operation: The position of all obstacles (Pipes) and the coordinates of the bird are fixed to their initial values.

### b. PLAYING: The actual game is active.

- Operation: Gravity logic is applied to lower the bird, and the vertical speed is updated each time a mouse-click signal is received. The pipe moves to the left, and the score counter is activated.

### c. PAUSED:

- Description: Enter when the user presses a certain button (KEY2) during the game.
- Operation: All coordinate updates by frame_tick are interrupted, leaving the screen frozen.

### d. GAME_OVER:

- Description: Enter when the bird hits the pipe or hits the ground (bird_dead = '1).
- Operation: Display "GAME OVER" with large text (scaling applied) on the screen and secure the final score to the HEX display.

## ii. State Transition Logic

- IDLE → PLAYING: Toggle when the user presses the Start button (SW0).
- PLAYING → PAUSED: Toggle when pressing the Pause button (KEY2) during the game.
- PAUSED → PLAYING: Press the Pause button again to resume the game.
- PLAYING → GAME_OVER: Switching when the collision_event signal is generated from the collision detection module.
- GAME_OVER → IDLE: Return to the initial screen after a reset signal has been entered or a certain period of time.